An explicit capture energy for a critical polarized Duffing map

Abstract

We give an explicit phase energy for the critical polarization of a linearly implicit two-step Duffing scheme. The inherited polarized energy controls the product of the two positions and both momenta, but its sublevels are noncompact. A corrected position imbalance supplies the missing control. On each fixed polarized-energy cap, the absolute imbalance has a one-sided increment bound obtained before either position is known to be bounded. Adding it to the inherited energy gives compact invariant sublevels for every positive step and damping parameter and every independent initial pair. A sum over two phase iterates is strictly decreasing outside the full stationary-pair set. Finite variation of the imbalance then selects one stationary pair; in the original indices, distinct even and odd limits are possible. The recurrence, polarized dissipation and finite-step Lyapunov principles have established antecedents. The result is confined to this critical map, and close-source access limitations are stated explicitly.

Keywords: Lyapunov function; polarized energy; boundedness; stationary pairs

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1 The recurrence and the phase energy

Consider the linearly implicit recurrence

$$\begin{aligned} \frac{q_{j+1}-q_{j-1}}{2h} &= \frac{p_{j+1}+p_{j-1}}{2}, \\ \frac{p_{j+1}-p_{j-1}}{2h} &= q_j - q_j^2\frac{q_{j+1}+q_{j-1}}{2} - d\frac{p_{j+1}+p_{j-1}}{2}, \qquad j \ge 1, \end{aligned} \tag{1}$$

where $h, d > 0$ are fixed and $x_0 = (q_0, p_0), x_1 = (q_1, p_1) \in \mathbb{R}^2$ are independent initial states. This is equation (7) of Sato, Matsuo and Furihata [1] at the polarization parameters $a = 1, b = 0$. It discretizes the damped Duffing equation, but the relevant state for its autonomous dynamics is a pair of successive original states.

For $s = (q, p, Q, P)$ define

$$\begin{aligned} \sigma(s) &= \frac{2[p + hQ(1-Qq)]}{1 + hd + h^2Q^2}, \qquad q' = q + h\sigma(s), \quad p' = \sigma(s) - p, \\ \tau(s) &= \frac{2[P + hq'(1-q'Q)]}{1 + hd + h^2(q')^2}, \quad Q' = Q + h\tau(s), \quad P' = \tau(s) - P, \\ Ts &= (q', p', Q', P'). \end{aligned} \tag{2}$$

Thus $s_n = T^n s_0 = (x_{2n}, x_{2n+1})$ advances two original updates at a time. Primes in (2) denote this single phase update. Set

$$\begin{aligned} H(s) &= (qQ-1)^2 + p^2 + P^2, \qquad g(u) = u - u^2, \\ M(s) &= q^2 - Q^2 + \frac{2}{d}(qp - QP) + \frac{2h}{d} g(qQ), \\ S(s) &= \sigma(s)^2 + \tau(s)^2. \end{aligned} \tag{3}$$

Here H is a shifted and scaled polarized pair energy. It is not the mechanical Duffing energy of one original state.

For every fixed finite $B \geq 0$, write $\Omega_B = \{H \leq B\}$ and define

$$\begin{aligned} &K = 1 + \sqrt{B}, \qquad C_0 = \frac{2h}{d}(K + K^2), \quad D_0 = \frac{2B}{d^2} + C_0, \\ &L_0 = 2D_0 + 2K, \quad L_1 = 2L_0 + 8h^2 B, \quad C_B = \frac{h}{d}(1 + h^2 L_1), \\ &\kappa_B = \frac{hd}{2C_B}. \end{aligned} \tag{4}$$

All these constants remain frozen along an orbit. The energies and the norm bound used below are

$$\begin{aligned} V_B(s) &= H(s) + \kappa_B |M(s)|, \\ Z_B(s) &= V_B(s) + V_B(Ts), \\ R_B(A) &= \frac{2A}{\kappa_B} + 4K + 2D_0 + B \qquad (A \geq 0). \end{aligned} \tag{5}$$

These are formulas in the actual four-dimensional phase. The composition in Z_B uses one explicitly known application of (2); no history or first-entry time enters the definition.

Theorem 1. *For every fixed $h, d > 0$, recurrence (1) is uniquely defined from every independent pair $x_0, x_1 \in \mathbb{R}^2$. The map T is smooth on $\mathbb{R}^4$. For each finite $B \geq 0$, the following assertions hold.*

For every $s \in \Omega_B$,

$$H(Ts) - H(s) = -hdS(s), \qquad V_B(Ts) - V_B(s) \leq -\frac{hd}{2} S(s). \tag{6}$$

For every finite $A \geq 0$, the set $\{H \leq B, V_B \leq A\}$ is compact and positively T-invariant, and

$$\|s\|^2 \leq R_B(A) \qquad (H(s) \leq B,\ V_B(s) \leq A). \tag{7}$$

In particular, choosing $B = H(s_0)$ and $A = V_B(s_0)$ bounds all original states.

The full fixed set is

$$\mathcal{F} = \mathrm{Fix}(T) = \{(0,0,0,0)\} \cup \{(q,0,Q,0) : qQ = 1\}. \tag{8}$$

Moreover, for $s \in \Omega_B$, with $\mathcal{L} = S + S \circ T$,

$$Z_B(Ts) - Z_B(s) \leq -\frac{hd}{2} \mathcal{L}(s), \qquad \mathcal{L}(s) = 0 \iff s \in \mathcal{F}. \tag{9}$$

Thus Z_B is strictly decreasing on $\Omega_B \setminus \mathcal{F}$ and is unchanged on $\mathcal{F} \cap \Omega_B$. Its sets $\{H \le B, Z_B \le A\}$ are also compact and positively invariant and satisfy (7).

For every s_0, put $B = H(s_0)$, $A = V_B(s_0)$ and $R = R_B(A)$. Then

$$\sum_{n=0}^{\infty} S(s_n) \le \frac{H(s_0)}{hd}, \qquad \sum_{n=0}^{\infty} |M(s_{n+1}) - M(s_n)| \le \frac{(1+h^2R)H(s_0)}{d^2}. \tag{10}$$

Write $m = \lim_n M(s_n)$. The phase s_n converges to one member of $\mathcal{F}$. Either $H(s_n) \to 1$ and $s_n \to 0$, or $H(s_n) \to 0$ and, for some $\varepsilon \in \{-1, 1\}$,

$$s_n \longrightarrow (\varepsilon r, 0, \varepsilon/r, 0), \qquad r = \left(\frac{\sqrt{m^2+4}+m}{2}\right)^{1/2} > 0. \tag{11}$$

The theorem imposes no upper bound on h. Its energy coefficients depend on the fixed cap B, and it asserts capture of each orbit in a compact invariant set, rather than a common compact absorber. Such a common absorber is impossible because the stationary-pair hyperbola in (8) is unbounded. The strict energy is relative to this entire set; positive definiteness at one prescribed equilibrium and stability of each stationary pair are not asserted.

In the original indices, (11) reads

$$x_{2n} \longrightarrow (\varepsilon r, 0), \qquad x_{2n+1} \longrightarrow (\varepsilon/r, 0). \tag{12}$$

These limits form a genuine period-two solution of (1) when $r \neq 1$. There is no assertion that every original sequence converges to a single physical Duffing equilibrium.

The inherited method and energy belong to the polarized discrete-gradient setting of [1–3]. In the complete published Duffing analysis [1, Lemma 8 and Proposition 11], compact polarized-energy sublevels and convergence to true Duffing equilibria are obtained in the strict range $a > 1, b < 1/2$. At $a = 1, b = 0$, the squared position-difference term in that energy vanishes. The purpose of the correction in (3) is to control this missing direction from the actual updates. The finite-sum conversion in (5) and the compact limit-set argument are standard Lyapunov devices. The source's semi-strong equality mechanism is already given in [1, Lemma 10]; the present proof writes the relevant zero set directly and then resolves its non-discrete stationary-pair geometry. Section 5 states the remaining source-access qualifications.

2 Exact loss and imbalance identities

Lemma 1. *The formulas* (2) *give the unique two successive updates of* (1). *For $s_{1/2} = (q', p', Q, P)$,*

$$H(s_{1/2}) - H(s) = -hd\sigma(s)^2, \qquad H(Ts) - H(s_{1/2}) = -hd\tau(s)^2. \tag{13}$$

On all of $\mathbb{R}^4$,

$$M(Ts) - M(s) = \frac{h}{d}\left[(1+h^2Q^2)\sigma(s)^2 - (1+h^2(q')^2)\tau(s)^2\right]. \tag{14}$$

Proof. Write $u = \sigma(s)$ and $v = \tau(s)$. The first original update requires

$$q' - q = h(p' + p), \qquad p' - p = h\{2Q - Q^2(q' + q) - d(p' + p)\}. \tag{15}$$

Putting $u = p' + p$ gives

$$(1 + hd + h^2Q^2)u = 2[p + hQ(1 - Qq)].$$

The coefficient is positive, so this gives a unique update and the first line of (2). The second update has the same argument with the middle position q'. All denominators are positive, proving global smoothness as well.

The first energy change is

$$\begin{aligned} H(s_{1/2}) - H(s) &= huQ[Q(q' + q) - 2] + u(p' - p) \\ &= huQ[Q(q' + q) - 2] + hu[2Q - Q^2(q' + q) - du] = -hdu^2. \end{aligned}$$

The second identity in (13) follows in the same way. In particular, s, $s_{1/2}$ and Ts belong to the same cap whenever $s \in \Omega_B$.

For the imbalance, put $f(z, w) = z^2 + (2/d)zw$ and $\bar{q} = (q + q')/2$. Product differences and (15) give

$$f(q', p') - f(q, p) = \frac{4h}{d} g(\bar{q}Q) + \frac{h}{d}u^2.$$

Since g is quadratic,

$$4g((\xi + \eta)/2) = 2g(\xi) + 2g(\eta) + (\eta - \xi)^2.$$

Use $\xi = qQ$, $\eta = q'Q$ and $\eta - \xi = hQu$ to obtain

$$f(q', p') - f(q, p) = \frac{2h}{d}[g(qQ) + g(q'Q)] + \frac{h}{d}(1 + h^2Q^2)u^2. \tag{16}$$

The corresponding second update gives

$$f(Q', P') - f(Q, P) = \frac{2h}{d}[g(q'Q) + g(q'Q')] + \frac{h}{d}(1 + h^2(q')^2)v^2.$$

Subtracting these formulas and adding the change of $(2h/d)g(qQ)$ cancels all the g terms. This is (14). ∎

3 Capture from the noncompact energy cap

Lemma 2. *For $s \in \Omega_B$ the following estimates hold without a prior position bound:*

$$\begin{aligned} |qQ| &\le K, & p^2 + P^2 &\le B, \\ |M(s) - (q^2 - Q^2)| &\le \frac{q^2 + Q^2}{2} + D_0, & & \\ q^2 + Q^2 &\le 2|M(s)| + 4K + 2D_0. & & \end{aligned} \tag{17}$$

The sign of M localizes one coordinate:

$$M(s) \ge 0 \implies Q^2 \le L_0, \qquad M(s) \le 0 \implies q^2 \le L_0. \tag{18}$$

Finally,

$$|M(Ts)| - |M(s)| \le C_B S(s). \tag{19}$$

Proof. The first line of (17) follows from the definition of H. Let $X = q^2 + Q^2$. The remaining part of M satisfies

$$\begin{aligned} |M - (q^2 - Q^2)| &\le \frac{2}{d}\sqrt{X}\sqrt{B} + \frac{2h}{d}(K + K^2) \\ &\le \frac{X}{2} + \frac{2B}{d^2} + C_0 = \frac{X}{2} + D_0. \end{aligned}$$

Here we used Cauchy–Schwarz and the elementary square inequality. Also $\min(q^2, Q^2) \le |qQ| \le K$, whence

$$|q^2 - Q^2| \ge X - 2K.$$

Combining these two estimates gives the last line of (17).

To prove (18), the preceding remainder estimate gives

$$\frac{1}{2}q^2 - \frac{3}{2}Q^2 - D_0 \le M \le \frac{3}{2}q^2 - \frac{1}{2}Q^2 + D_0.$$

If $Q^2 > L_0 = 2D_0 + 2K$, then $Q^2 > 2K$ and $q^2 \le K^2/Q^2 < K/2$. The upper bound makes $M < -K/4 < 0$. This proves the first implication by contradiction. Interchanging the two large-coordinate cases proves the second implication. Neither argument bounds the whole position pair.

By (13), every intermediate momentum has magnitude at most $\sqrt{B}$. Therefore

$$\sigma(s)^2 = (p + p')^2 \le 4B, \qquad \tau(s)^2 = (P + P')^2 \le 4B. \tag{20}$$

Put $m = M(s)$ and $m' = M(Ts)$. If $m' = 0$, then $|m'| - |m| \le 0$. If $m' > 0$, then

$$|m'| - |m| \le m' - m.$$

Apply (18) at Ts, which is in the same cap by Lemma 1. It gives $(Q')^2 \le L_0$, so

$$Q^2 = (Q' - h\tau)^2 \le 2L_0 + 2h^2\tau^2 \le 2L_0 + 8h^2B = L_1.$$

In (14), discard the nonpositive τ^2 term. It follows that

$$|m'| - |m| \le \frac{h}{d}(1 + h^2Q^2)\sigma^2 \le C_B\sigma^2.$$

If $m' < 0$, then $|m'| - |m| \le -(m' - m)$, and (18) at Ts gives $(q')^2 \le L_0 \le L_1$. Negating (14) and discarding the nonpositive σ^2 term yields

$$|m'| - |m| \le \frac{h}{d}(1 + h^2(q')^2)\tau^2 \le C_B\tau^2.$$

These three endpoint cases include both sign crossings and all zero endpoint values. They prove (19) on the entire cap using only the cap invariance already established. ∎

Proof of capture in Theorem 1. Combining (13) and (19), with $\kappa_B C_B = hd/2$, proves (6). If $H \le B$ and $V_B \le A$, nonnegativity gives $|M| \le A/\kappa_B$. The last line of (17) and $p^2 + P^2 \le B$ then give exactly (7).

The set $\{H \le B, V_B \le A\}$ is closed because its defining functions are continuous, and bounded by (7). It is compact in $\mathbb{R}^4$. Both defining functions are nonincreasing under T, so the set is positively invariant. For any finite initial phase choose $B = H(s_0)$ and $A = V_B(s_0)$. This captures every full phase before bounded positions are used anywhere in the proof. Every original state is a component of a full phase; the intermediate phase combines components of consecutive captured phases and is bounded too.

At $B = 0$, the cap already forces $p = P = 0$ and $qQ = 1$. These are stationary phases, but the constants in (4) remain finite with $C_B, \kappa_B > 0$, and (7) remains valid. No zero-coordinate exception or positive-energy assumption is needed. ∎

4 Strictification and selection of one pair

Lemma 3. *The fixed set is* (8)*, and* $S(s) + S(Ts) = 0$ *if and only if* $s \in \mathcal{F}$*.*

Proof. If $Ts = s$, the two position equalities in (2) imply $\sigma = \tau = 0$. The two momentum equalities then force $p = P = 0$, and the update numerators give

$$Q(1 - qQ) = q(1 - qQ) = 0.$$

Either $qQ = 1$, or $q = Q = 0$. Conversely, all these phases are fixed.

If $S(s) + S(Ts) = 0$, all four squared increments vanish. The first two give $Ts = (q, -p, Q, -P)$. The first-update zero numerators at s and Ts are

$$p + hQ(1 - qQ) = 0, \qquad -p + hQ(1 - qQ) = 0,$$

so $p = 0$ and $Q(1 - qQ) = 0$. Since both first increments vanish, the second updates use the same middle position q. Their zero numerators similarly give $P = 0$ and $q(1-qQ) = 0$. Thus $s \in \mathcal{F}$. The converse follows from stationarity. ∎

One phase block alone need not have a strict loss. For example, at $h = d = 1$, the phase $(0, -1, 1, 0)$ has $S = 0$ and maps to $(0, 1, 1, 0)$, which differs from it. This explains the use of the finite sum Z_B.

Proof of strictification and selection in Theorem 1. Applying (6) at s and Ts in the same cap gives

$$Z_B(Ts) - Z_B(s) = V_B(T^2 s) - V_B(s) \le -\frac{hd}{2}[S(s) + S(Ts)].$$

Lemma 3 proves the strictness assertion. Since $Z_B \ge V_B$, its cap sublevels are bounded by the same $R_B(A)$; continuity and the decrement prove compactness and positive invariance.

Fix now an arbitrary initial phase and the constants stated before (10). Telescoping the exact H loss and using $H \ge 0$ gives the first sum in (10). In particular, $\sigma(s_n), \tau(s_n) \to 0$. In (14) at s_n, the position Q is a coordinate of s_n and q' is a coordinate of s_{n+1}. Capture has already bounded both squared coordinates by R. Hence

$$|M(s_{n+1}) - M(s_n)| \le \frac{h}{d}(1 + h^2 R) S(s_n).$$

Summing proves the second bound in (10), and thus $M(s_n) \to m$ for some finite m.

The nonincreasing energy has a limit $H(s_n) \to c \ge 0$. Compactness supplies cluster points. Along any subsequence converging to z, continuity and $S(s_n) + S(s_{n+1}) \to 0$ imply $\mathcal{L}(z) = 0$. By Lemma 3, every cluster point is in $\mathcal{F}$. The continuous H has value 1 at the origin and 0 everywhere else in $\mathcal{F}$, so $c \in \{0, 1\}$.

If $c = 1$, the origin is the only possible cluster point. Compactness then proves convergence of the whole phase sequence to it. If $c = 0$, a cluster point satisfies $p = P = 0$, $qQ = 1$, and $q^2 - Q^2 = m$. These equations leave precisely the two points

$$(r, 0, 1/r, 0), \quad (-r, 0, -1/r, 0), \qquad r^2 = \frac{\sqrt{m^2 + 4} + m}{2} > 0.$$

It remains to choose one of them for the actual sequence. Write $s_n = (\widehat{q}_n, \widehat{p}_n, \widehat{Q}_n, \widehat{P}_n)$, so hats index full phase iterates rather than original updates. For all sufficiently large n, $H(s_n) < 1$. The intermediate loss also gives $H(s_{n+1/2}) < 1$. Consequently,

$$\widehat{q}_n \widehat{Q}_n > 0, \qquad \widehat{q}_{n+1} \widehat{Q}_n > 0.$$

Thus $\widehat{q}_n, \widehat{Q}_n, \widehat{q}_{n+1}$ share the same nonzero sign for every sufficiently large n. One eventual sign ε excludes the other point of the fiber. Compactness with a single remaining cluster point proves (11), and the phase convention gives (12). ∎

5 Normalization, source scope and verification

For this critical polarization the displayed pair energy in [1] is

$$\widehat{G}((q, p), (Q, P)) = \frac{p^2 + P^2}{4} - \frac{qQ}{2} + \frac{q^2 Q^2}{4}, \qquad H = 4\widehat{G} + 1. \tag{21}$$

On the diagonal it reduces to $p^2/2 - q^2/2 + q^4/4$. For the displayed polarized discrete gradient $D(x, y, z)$ at $a = 1, b = 0$, direct expansion gives

$$\widehat{G}(x, y) - \widehat{G}(y, z) = \tfrac{1}{2} D(x, y, z) \cdot (x - z).$$

With the literal recurrence $(x - z)/(2h) = JD$ and $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & -d \end{smallmatrix}\right)$, the energy quotient is therefore $D^T JD$, whereas equation (8) of the published paper prints $2D^T JD$. The exact losses used here are derived from (1) in Lemma 1. This positive-factor normalization discrepancy leaves the source's dissipation sign and zero-loss mechanism unchanged. A rational witness and the universal chain identity are included in the companion checks.

The complete published four-page body of [1] and its METR2015-04 author report were compared. The final Lemma 8 and Proposition 11 retain the strict assumptions $a > 1, b < 1/2$; Lemma 7 excludes $a = 1$ in matching discrete stationary pairs with physical equilibria. Its Lemma 10 supplies the relevant finite-window equality mechanism. Our proof does not apply the source's compact-sublevel conclusion at a parameter where that hypothesis is absent, and it does not apply a discrete-fixed-set convergence criterion to the hyperbola (8).

The complete scalar predecessor [2] and the complete quartic-stabilization comparison [4] were not available for full-body inspection. Their metadata and abstracts, together with the predecessor's description in [1], do not exclude a related balance argument or an overlapping endpoint result. This is a material limitation on priority and coverage assessment. No firstness or presently open-problem claim is made. The cited polarization framework [3] supplies conservation, consistency and linear-implicitness results in its inspected sections; those results alone do not give capture for the noncoercive damped pair energy here.

The reader package contains a standalone exact checker, deterministic output and a technical verification guide. The checker corroborates the loss and imbalance identities, cap estimates, zero and sign-crossing cases, the strict zero set and the norm/variation constants using symbolic and rational arithmetic. The universal statements, compactness and infinite-time selection rest on the analytic proof above, rather than a finite set of trajectories. The result concerns only $a = 1, b = 0$; no larger polarization class, optimal coefficient, convergence rate or basin description is inferred.

References

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